

Bhatiya Kavya Vishnukumar

Bachelor of Technology in Artificial Intelligence
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EDUCATION

•Bachelor of Technology in Artificial Intelligence

2024 – 2028

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat

CGPA: 9.05 / 10.00 (till 4th Semester)

PROJECTS

•LogiFlow — Multimodal AI Logistics Optimization Platform

FastAPI, PostgreSQL, Next.js, Gemini API, CatBoost

- Built AI-powered multimodal logistics decision intelligence platform supporting road, rail, air, and sea route optimization.
- Developed FastAPI backend integrating traffic analysis, demand forecasting, asset utilization monitoring, delay prediction, and risk-aware route ranking pipelines.
- Containerized backend services with Docker and deployed FastAPI workloads on Google Cloud Run.
- Built and deployed a delay-risk prediction service using HistGradientBoostingClassifier for proactive shipment risk assessment.
- Achieved **73.0% accuracy**, **68.0% delayed-shipment recall**, and **78.3% cross-validated ROC-AUC** using stratified holdout and 5-fold validation.
- Implemented multi-objective optimization and ranking pipelines using delivery time, reliability, cost, and sustainability constraints.
- Integrated Gemini-powered explainability engine for route recommendations; selected among **Top 106 teams globally** in Google Solution Challenge 2026.

•AirHelp — AI Airport Assistance Platform

FastAPI, Gemma, Next.js

- Built a multilingual AI airport assistance platform supporting real-time passenger query resolution and navigation assistance.
- Developed FastAPI backend services and orchestration pipelines for speech processing, context retrieval, and response generation.
- Designed modular LLM integration workflows enabling scalable conversational assistance across multiple airport service scenarios.

•MindSettler — Counseling Session Management Platform

Django REST Framework, SQL

- Architected a role-based backend system for appointment scheduling, counselor approvals, and session lifecycle management.
- Developed RESTful APIs and normalized SQL schemas to support scalable and consistent data operations.
- Implemented authentication, authorization, and workflow automation with maintainable service-layer abstractions.

ENGINEERING RESEARCH

•SMSRF — Multi-Stakeholder Recommendation Framework

Python, NumPy, pymoo, NSGA-II, MovieLens 100K, MovieLens 1M

- Designed and implemented a multi-objective recommendation engine using NSGA-II to optimize Accuracy, Novelty, Diversity, and Serenity objectives.
- Developed reusable optimization, evaluation, and Pareto-selection pipelines for large-scale recommendation experiments.
- Built a configurable experimentation framework supporting metric computation, policy evaluation, and comparative benchmarking.
- Evaluated system behavior on **MovieLens 100K** and reproduced similar trends on **MovieLens 1M**, improving result reliability and scalability.
- Achieved up to **12.6% higher Novelty** and **20.9% higher Serenity** than NSGA-3 baselines.
- Engineered reproducible research workflows with statistical validation, robustness analysis, and automated experiment tracking.

ACHIEVEMENTS

- Selected among **Top 106 teams globally** in Google Solution Challenge 2026 for LogiFlow.
- **LeetCode**: 430+ problems solved | Max Contest Rating: 1620.
- Finalist, **Web Wonders 2025**, NIT Surat.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, C

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, DBMS

Backend: FastAPI, Django REST Framework, Express.js, REST APIs

AI/Optimization: Scikit-learn, CatBoost, Recommendation Systems, NSGA-II, pymoo

Databases: SQL, PostgreSQL, MongoDB

Data: NumPy, Pandas

Tools: Git, GitHub, Docker, Cursor, Hugging Face Spaces